

IoT enabled E-Vehicle Charging System with Battery Monitoring and Charge Scheduling

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Abstract—The prominence of Electric Vehicles (EVs) is increasing significantly due to their ability to drastically reduce carbon-dioxide and greenhouse gas emissions from road transportation, contributing to the fight against global warming. However, the limited availability of charging stations and the time required for charging have been inhibiting factors for potential buyers to adopt EVs. As a result, there is a need for a smart charging system to reduce the challenges of locating charging station and checking slot availability. The primary objective of the project is to design and implement an EV charger featuring a rectifier circuit with Power Factor Correction (PFC) controller using a flyback converter topology. Additionally, a web application for EV charging station “ChargEVfy” has been developed. This application aims to assist EV drivers in identifying nearby charging stations. Once a charging station is located, users have an option to check available charging slots and even reserve a slot in advance for their vehicle by registering on “ChargEVfy” website. The web application provides an option whereby the users can input their starting point and destination. Subsequently, the webpage displays a roadmap with all accessible charging stations along the journey. To enable these functionalities, the proposed system is equipped with IR sensors, Arduino Nano, ESP8266 Wi-Fi module, Blynk IoT cloud server and web application development tools.

Keywords— *Electric Vehicle (EV), charging station, battery monitoring, slot booking, Internet of Things (IoT)*

I. INTRODUCTION

New energy cars have advantages over conventional vehicles in terms of reduced emissions, high energy economy, and cheap operating costs. Compared to a comparable petrol or diesel vehicle, an electric vehicle has significantly reduced operating costs. Instead of utilizing fossil fuels like petrol or diesel to charge their batteries, electric vehicles use electricity. Due to their greater efficiency and the lower cost of power, charging an electric car is more affordable than purchasing fuel or diesel for the travel needs.[1]

By 2030, there will be 45–50 million electric vehicles on Indian roads, giving EV charging companies a sizable market opportunity. There is a demand for sustainable and effective electric vehicle charging with the rise in the number of electric vehicles. Infrastructure is required for a decentralized mobile app to locate charging stations and provide accurate information to the user about location. [2] To set up an EV charging station, the location of the charging station is a major challenge.

Location or design of charging station should be such that it is easily visible, accessible, save time, and charging queue can be minimized. This provided the motivation for this research.

The main contributions of this paper can be summarized as follows: Design of a 3.3 kW charger circuit which includes a rectifier and PFC controller, along with flyback converter for charging the E-Vehicles, developing a web application named “ChargEVfy” which acts as a user interface between the charging station and the EV drivers, and design and implementation an IoT integrated system was carried out, to (i) locate the nearby charging stations, (ii) check for slot availability at the charging stations and enables booking a slot in advance at the webpage, (iii) monitor the battery parameters like voltage, current and temperature of the EV.

II. LITERATURE REVIEW

The feasibility of a flyback charger for electric vehicles based on silicon-carbide (SiC) devices is presented in [3]. The flyback topology is simplest, but its device voltage stresses are strongly affected by the transformer leakage inductance. To reduce the device voltage stresses by recovering the energy stored in the transformer leakage inductance, an active-clamp circuit and active snubber circuit are adopted on both sides of the transformer. Based on the analysis, a 3.3 kW/100 kHz prototype was designed and experimented to show the feasibility. The general review of power factor correction of single-phase AC/DC-DC power electronic converters for electric vehicle applications is discussed [4]. To transfer power between the grid and EV unidirectional charger system is implemented for improved power quality. IoT-Based Electrical Vehicle’s Energy Management and Monitoring System is proposed in [5]. The proposed work consists of battery management and monitoring system of electric vehicles, low-cost and IoT-based, in real-time, and easily used to help users through an application supporting the Internet of Things technology to display the essential information required about the battery’s status such as battery capacity and the consuming current. This information is updated and displayed in real-time. The proposed system was implemented using an ESP32 microcontroller, Blynk mobile application, and Blynk IoT platform. IoT based slot locator in charging stations as proposed in [6] has given a solution to one of the biggest challenges in the charging process of electric vehicles which is finding slot availability. This work necessitated the IoT and cloud computing. With NODEMCU as the controller,

this work illustrated the use of indicators to send signals whether the slot was available or not. For storing the resources and information Blynk cloud has been used. Moreover, an IoT based intelligent smart controllers for Electric vehicles proposed in [7] gives a solution to the one of the problems with EVs, which is the limited capacity of battery and charging infrastructure availability. This work consists of different types of sensors such as temperature, voltage, current and speed sensors. With the help of the sensors the state of charge of the vehicle is estimated and it is processed by microcontroller and stored in the dashboard. When the internet is available this information is passed to the cloud and abnormalities are detected. The various IoT based Electric Vehicle Charging and Parking system methods are reviewed in [8]. The various smart parking, charging and combined charging-parking system, to solve various issues related with it were compared. The various types of methods and techniques used for parking and charging are discussed. Various sensors, controllers, software and cloud servers in the market helped to make system automatic, reliable and user friendly along with development of efficient IoT platform.

While researchers have introduced a number of charging strategies and battery monitoring systems, none of the existing work presents an integrated solution that considers the challenges of locating charging stations and checking for slot availability and a smart charging system with EV battery

monitoring. In this work, we provide an entire integrated EV charging system that enables charging of an EV and monitors the EV's battery parameters, as well as providing EV users with the option of checking slot availability at the charging station and prebooking them.

III. PROPOSED SYSTEM

The proposed system is developed for the Integration of IoT and with respect to the EV charging system the power converter topologies are designed [9].

The overall block diagram of EV charging system integrated with IoT is given in Fig.1.

The main components in the proposed system involved are a diode rectifier with PFC control, followed by a power converter circuit to charge the EV battery. To monitor the battery level, current and voltage sensors are used whose signals are passed through ESP 32 micro controller and the data is stored in Blynk IoT cloud. Also the availability of free space is detected by IR sensors. These sensor data is sent to Arduino Nano which collects and processes it.

Further all these sensors data are sent and stored in Blynk IoT cloud using ESP8266 WiFi Module and monitored in the web application "ChargEVfy" where the user can monitor all the information through IoT and Pre-book a slot in advance depending upon the slot availability.

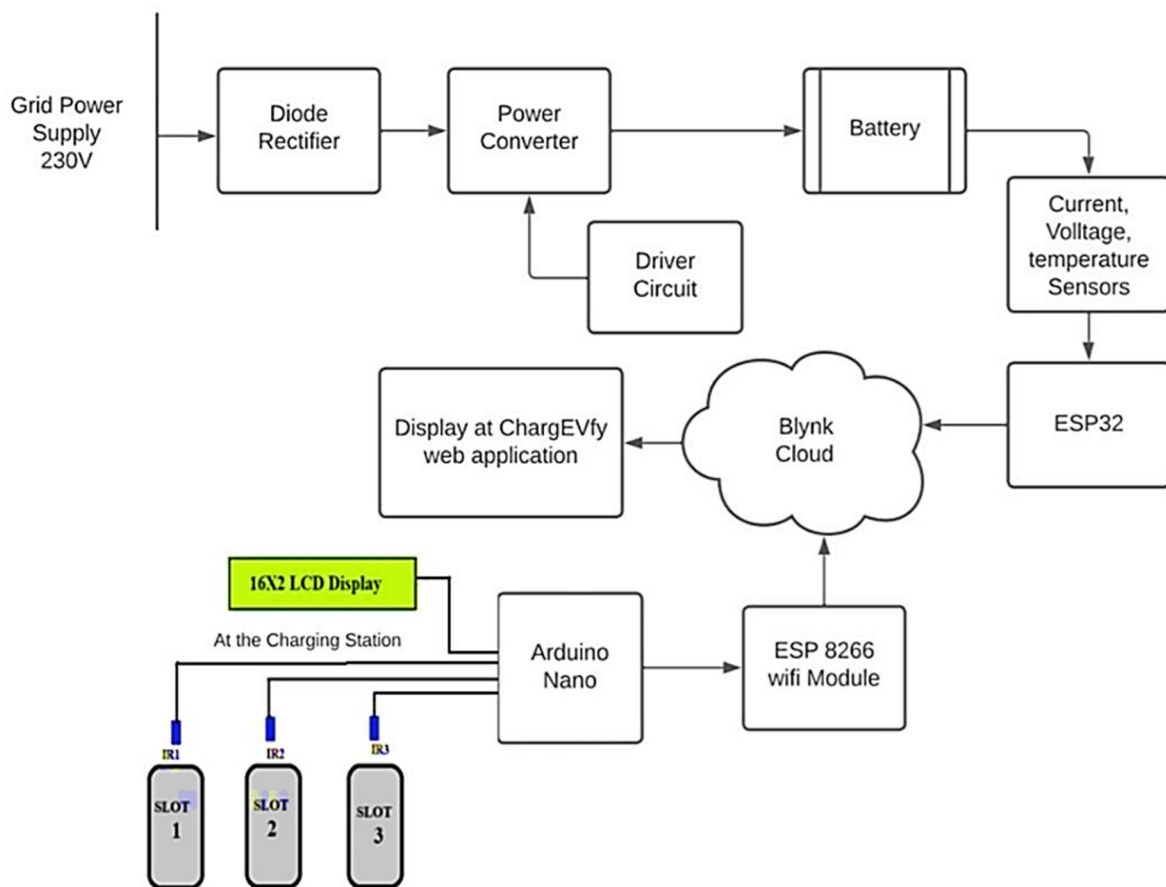


Fig. 1. Block Diagram of EV Charging System Integrated with IoT

IV. EV CHARGER CIRCUIT

As shown in the Fig.2, the EV battery charger circuit consists of input single phase AC supply of 230V, which is rectified with the help of a full bridge rectifier along with a PFC Boost converter circuit. Later this voltage is stepped down to meet the required battery voltage with the help of Flyback converter. This isolated buck-boost converter provides the required DC voltage to the EV's battery.

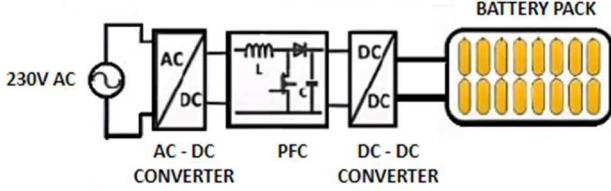


Fig. 2. EV Battery Charger Circuit

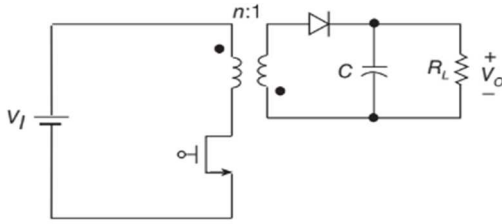


Fig. 3. Flyback Non-inverting Converter with the gate driven with respect to ground

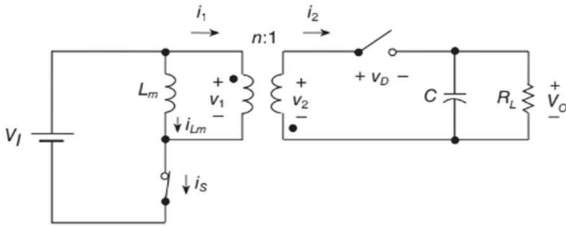


Fig. 4. Equivalent circuit when the switch is ON and the diode is OFF

A. FLYBACK CONVERTER

A flyback dc-dc converter given in Fig. 3 is a transformer (or isolated) version of the buck-boost converter. A transformer is used to eliminate any direct (dc) electrical connection between the input and the output of the converter power stage.

There are two modes of operation of flyback converters which are mainly: CCM (continuous conduction mode) and DCM (discontinuous conduction mode). Fig.4 and Fig.5 show equivalent circuits of the non-inverting flyback converter for CCM when the switch is on and the diode is off, and when the switch is off and the diode is on, respectively. The principle of operation of the flyback converter is as follows :

During the time interval $0 < t \leq DT$, the switch is on and the diode is off as indicated in Fig. 4. The voltage across the diode is $-(V_1/n + V_O)$, which maintains the diode in the off-state. The voltage across the magnetizing inductance L_m is V_1 and gives rise to a linear increase of the magnetizing inductance current with a slope of V_1 / L_m .

where , V_1 - primary side voltage

V_2 - secondary side voltage
 n - turns ratio
 V_O - output voltage across load
 L_m - magnetizing inductance
 D - duty ratio

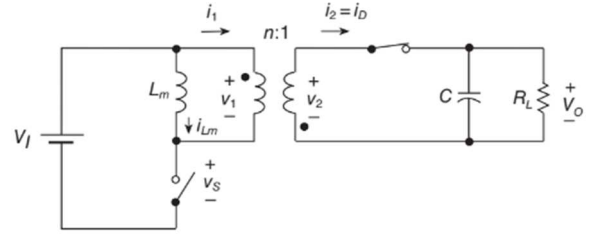


Fig. 5. Equivalent circuit when the switch is OFF and the diode is ON

During the time interval $DT < t \leq T$, the switch is off and the diode is on as shown in Figure 5. The voltage across the magnetizing inductance L_m is $-nV_O$, which causes the magnetizing inductance current to decrease linearly with a slope of $-nV_O/L_m$. The voltage across the switch is $V_1 + nV_O$. At time $t = T$, the switch is turned on again and the next cycle begins.

B. DESIGN OF FLYBACK CONVERTER

The flyback converter was designed for the E-bike (Hero Electric Photon-LP) whose specifications are mentioned in the Table I. The design is carried with the help of equations (1)-(13).

TABLE I. E-BIKE SPECIFICATIONS

Parameter Name	Values
Battery Capacity	72V
Current Rating	26Ah
Power Range	1.8 - 3.3 kW

The input grid voltage 230V AC acts as the source for the charger circuit which is rectified to DC with the help of a rectifier and PFC controller circuit. The boosted voltage of 400V from the PFC circuit is fed to the flyback converter. This output voltage of the PFC circuit acts as input voltage for the flyback converter.

TABLE II. FLYBACK CONVERTER SPECIFICATIONS

Parameter Name	Values
f_{sw}	100 kHz
V_{dc}	400V
V_o	72V
η	80%
V_r / V_o	1%
T	1×10^{-5} s

Here the input dc voltage V_{dc} of the converter is 400V and the required output voltage V_o is 72V. Assuming the voltage ripple $V_r / V_o \leq 1\%$, The maximum and minimum input dc

voltages are $V_{max} = 400$ V and $V_{min} = 350$ V, the maximum and minimum output currents are $I_{omax} = 25$ A and $I_{omin} = 46$ A respectively.

The maximum and the minimum output powers are given by,

$$P_{omax} = V_o \cdot I_{omax} \quad (1)$$

$$P_{omin} = V_o \cdot I_{omin} \quad (2)$$

The maximum and the minimum load resistances are given by,

$$R_{Lmin} = \frac{V_o}{I_{omax}} \quad (3)$$

$$R_{Lmax} = \frac{V_o}{I_{omin}} \quad (4)$$

The minimum and maximum dc voltage transfer functions are calculated as follows,

$$M_{VDCmin} = \frac{V_o}{V_{max}} \quad (5)$$

$$M_{VDCmax} = \frac{V_o}{V_{min}} \quad (6)$$

Let's assume efficiency $\eta = 85\%$ and the maximum duty ratio of $D_{max} = 0.4$, hence the transformer turns ratio n is calculated as,

$$n = \frac{\eta D_{max}}{(1 - D_{max}) M_{VDCmax}} \quad (7)$$

The minimum and the maximum duty cycle is given by,

$$D_{min} = \frac{n M_{VDCmin}}{M_{VDCmin} + \eta} \quad (8)$$

$$D_{max} = \frac{n M_{VDCmax}}{M_{VDCmax} + \eta} \quad (9)$$

Assuming the switching frequency $f_{sw} = 100$ kHz, the minimum magnetizing inductance is calculated as,

$$L_{m(min)} = \frac{n^2 R_{Lmax} (1 - D_{max})^2}{2 f_{sw}} \quad (10)$$

The peak to peak value of AC component of current through magnetizing inductance is

$$\Delta i_{Lm(max)} = \frac{n V_o (1 - D_{min})}{f_{sw} \cdot L_m} \quad (11)$$

The maxim input dc current with respect to V_{max} is given as

$$I_{Lmax} = M_{VDCmax} \cdot I_{omax} \quad (12)$$

$$C_{min} = \frac{D_{max} \cdot V_o}{f_{sw} \cdot R_{Lmin} \cdot V_{cgp}} \quad (13)$$

A list of circuit components used, and their ratings calculated from the design equations (1) - (13) are listed in Table III.

TABLE III. CALCULATED DESIGN PARAMETERS

Parameter Name	Values
n	3
P_{omax}	3300 W
D_{max}	0.414
$L_{m(min)}$	0.02 mH
C_{min}	265 μ F

C. SIMULATION DIAGRAM AND RESULTS

The proposed EV charger circuit is designed and simulated in MATLAB SIMULINK as shown in Fig.6. The PFC controller was designed as shown in the PFC controller block diagram as shown in Fig.7 using closed loop voltage and current control methods. PID (proportional integral derivative) controller was used with proportional and integral gain values $k_p = 2000$ and $k_i = 0.25$ which was found using trial and error method. The EV charger circuit was simulated in MATLAB Simulink and the following simulation results were obtained.

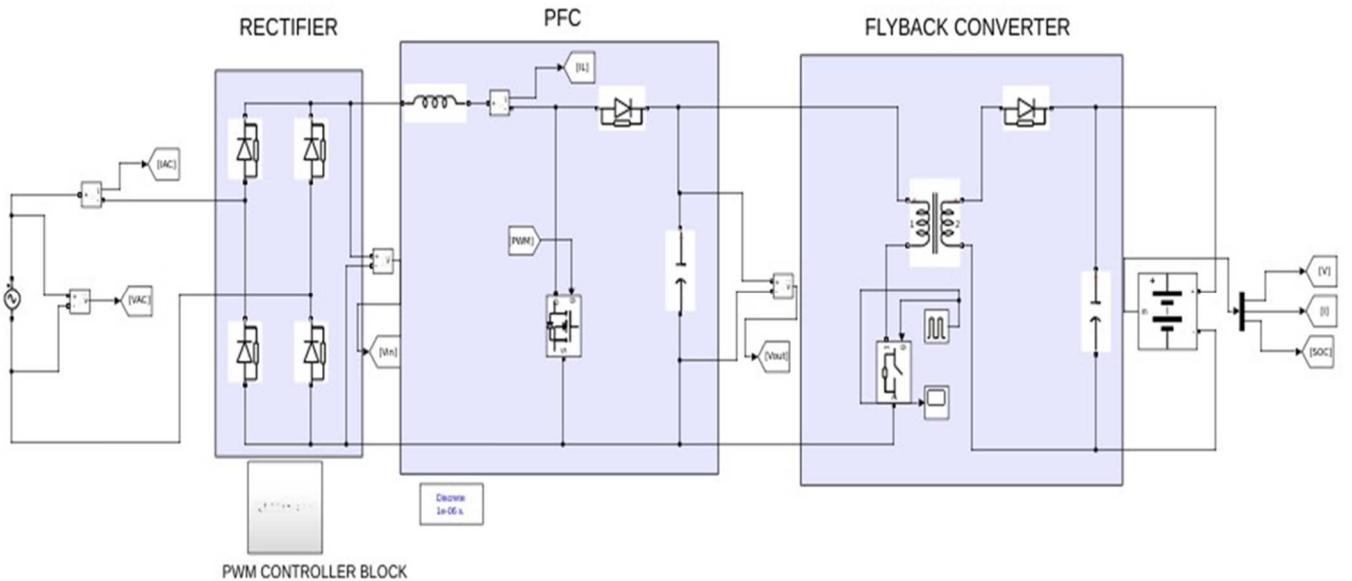


Fig. 6. Simulation Circuit Diagram of EV Charger

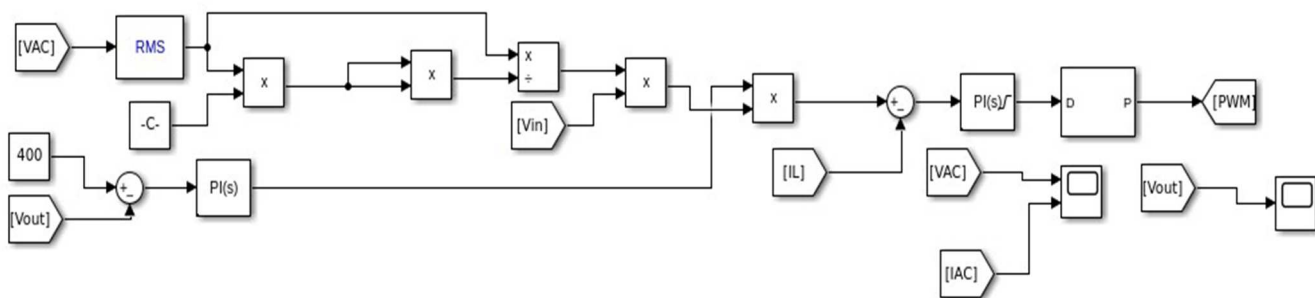


Fig. 7. Simulation Diagram of PFC Controller Block

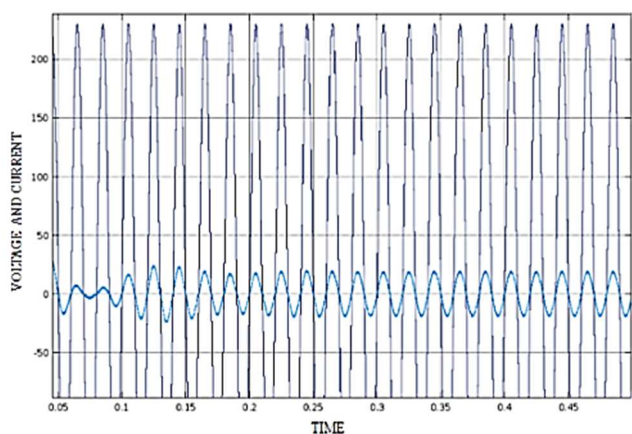


Fig. 8. Input side voltage and current waveforms

The voltage and current waveforms for the input supply was plotted as shown in Fig. 8. The closed loop PFC control has improved the power factor at supply side, as the voltage and current waveforms are in phase to each other showing the power factor is almost unity.

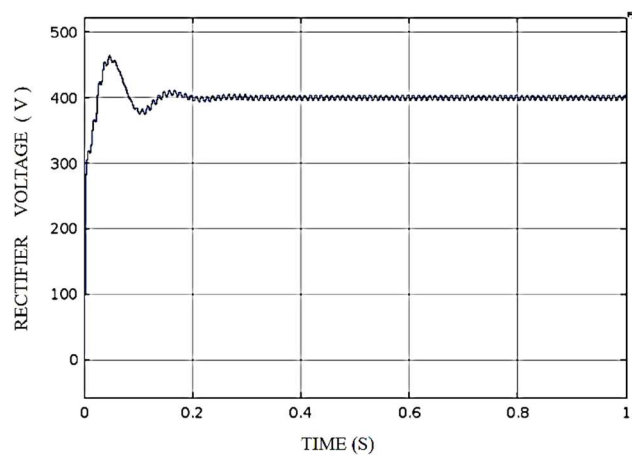


Fig. 9. PFC Rectified Output Voltage

It is observed that rectified output voltage of 400V with a lesser ripple was obtained at the end of rectified PFC boost circuit operation as shown in Fig.9. As shown in Fig. 10, a constant dc voltage of 71.6V was observed as the output battery voltage of flyback converter. Also, a dc current of 20A is obtained as the output battery current of flyback converter given in Fig. 11 and SOC is monitored for 69% as shown in Fig. 12.

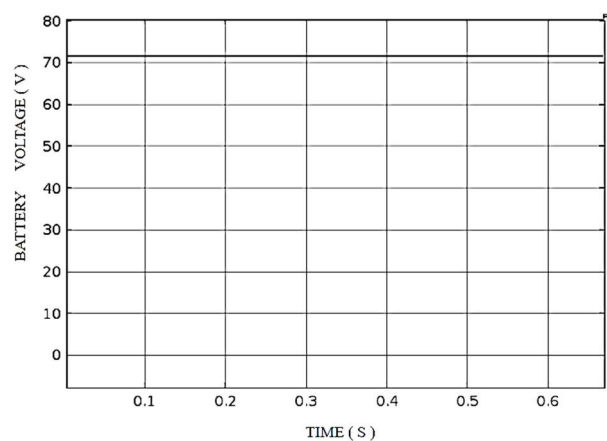


Fig. 10. Battery Output Voltage

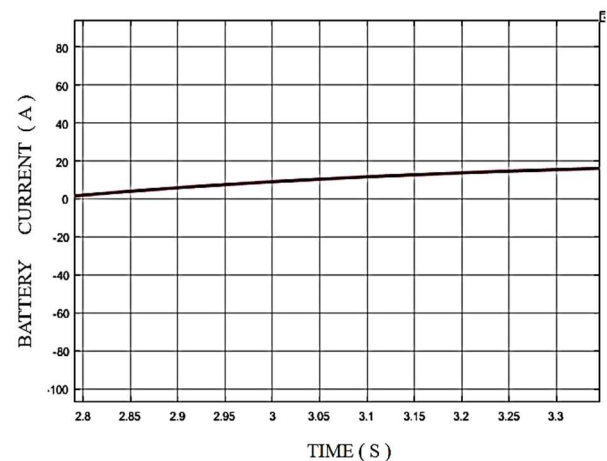


Fig. 11. Battery Output Current

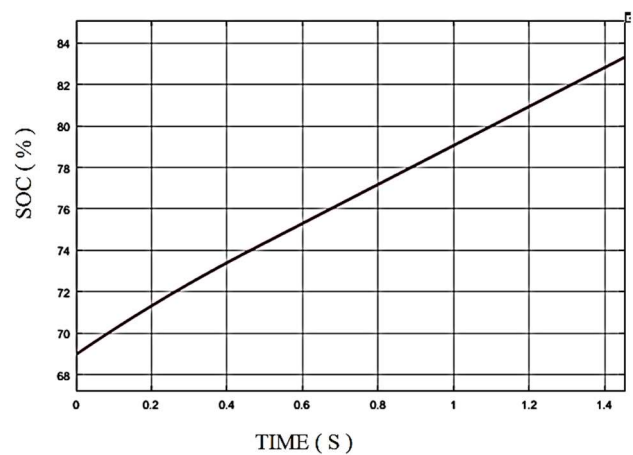


Fig. 12. SOC of Battery

V. BATTERY MONITORING SYSTEM (BMS)

The EV's battery management system is based on battery sensor technology. The battery's voltage, current, and temperature are continuously monitored by sensors such as voltage and current sensors. The ESP 32 microcontroller processes the data received from the sensors, such as the conversion between analog to digital, and transmits the processed data continuously to the cloud, where it is stored and accessible by internet-enabled devices including PCs, laptops, and Android smartphones with the right software. Internet access is necessary for two-way communication during EV charging.

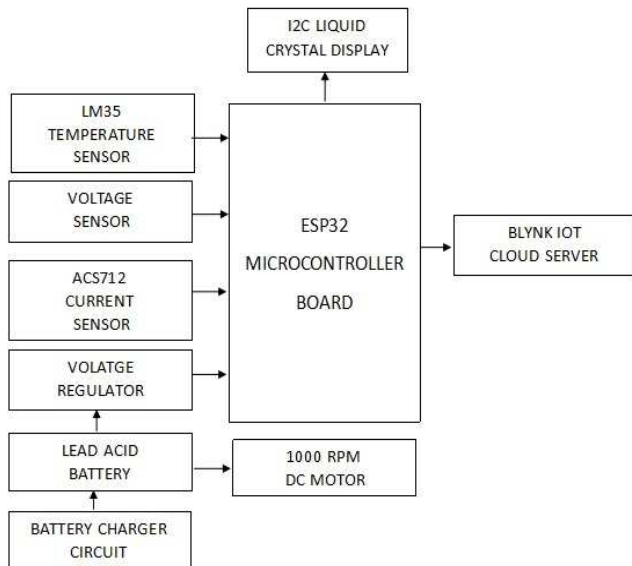


Fig. 13. Block Diagram of Battery Monitoring System using IoT

Fig.13 illustrates the block diagram of the proposed battery monitoring system. The 230V AC supply acts as input to the charger circuit (12 Volt 2A UPS Lead Acid Battery Charger circuit with auto cut off overcharge protection) which produces an output dc voltage of 14.5V, which is used to charge the 12V lead acid battery. Further a dc motor is connected to the battery which is controlled by a switch, it acts as a load to the circuit. IC 7805 and IC 7809 regulators are used to provide a regulated dc voltage of 5V and 9V respectively.

The ESP 32 is limited to 5V DC voltage input. In order to measure higher voltages we go with Voltage divider circuitry which typically consists of two resistors of 10k Ω connected in series. The temperature sensor attached to battery senses the temperature of the battery. It will sense the temperature of the battery and sends the sensed signal to the cloud using the ESP32 module. The current sensor is connected in series across the battery to measure the battery current.

The ESP 32 microcontroller converts analog data to digital data obtained from the sensors and continuously communicates the processed data to the Blynk IoT cloud. The voltage, current and temperature sensor values are displayed both at the LCD display and Blynk IoT Cloud dashboard simultaneously.

The list of various components used in the proposed battery monitoring system is shown in Table IV. Fig.14 shows the hardware setup of the battery monitoring system. This battery monitoring system enabled using IoT, has one-way communication to the web interface (ChargeVfy). Further the date from the vehicle to the cloud can also be monitored for safety related matters.

TABLE IV. COMPONENTS USED FOR BMS

Components Name	Specifications
Lead Acid battery	12V, 2Ah
Voltage Regulators	IC7805, IC7809
Temperature Sensor	LM 35
Current Sensor	ACS712
Voltage Sensor	10k Ω resistors
DC Motor	12V, 1000 RPM
Microcontroller Board	ESP 32 12E module
Charger Circuit	12V, 2A UPS
Cloud Server	Blynk IoT
LCD display	16x2 screen with I2C module

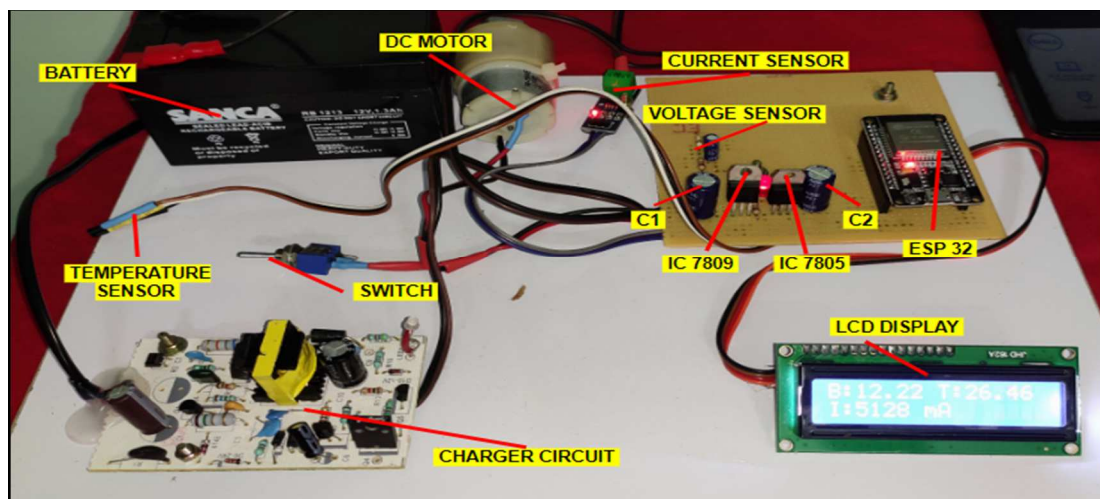


Fig. 14. Hardware Setup of Battery Monitoring System

VI. SLOT AVAILABILITY CHECKING USING IOT

Electric vehicles are more effective and have greater power delivery in addition to reducing pollution since they can use regenerative braking to replenish their batteries while moving. Even with all their benefits, electric vehicles still struggle with things like finding charging outlets. Unlike drivers of conventional vehicles, owners of EVs are unable to recharge their automobiles at any petrol station. Electric vehicle owners must keep their vehicles fully charged before leaving. There is no denying the necessity of building infrastructure like charging stations.

This EV Charging Station app is created using Flutter to assist EV drivers in finding available charging stations nearby. Users can reserve a time slot to charge their vehicle at a charging station after finding one. This technique can be used by EV owners to better effectively schedule their travels. The source and destination only need to be entered by the user. This technology creates a route with all accessible charging stations along the way based on these two characteristics. A user may also look for an EV station based on their location, city, or distance. By entering the source and destination together with the travel distance, the system gives a route with charging stations along the way.

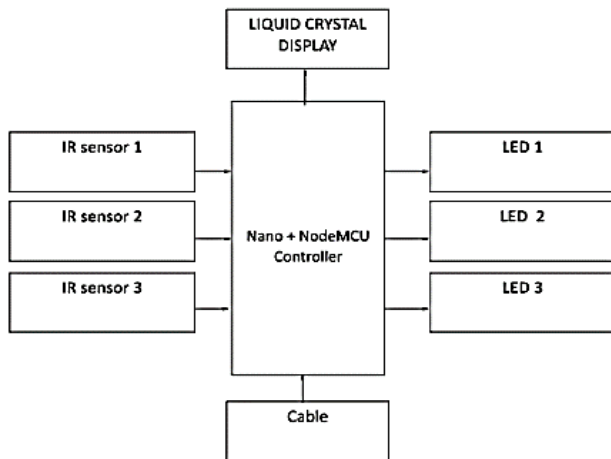


Fig. 15. Block diagram for Slot availability using IoT

Fig. 15 illustrates the components connection with the NODEMCU controller. It comprises three IR sensors, three

LEDs, a cable, a NODEMCU Esp8266, Arduino Nano, and an LCD display. An infrared sensor senses infrared radiation in its surroundings and produces an electric signal as a result. By aiming it at multiple points on the object being read, an IR sensor can provide more information throughout a measurement than a contact device. There is no data leakage owing to IR radiation beam directionality.

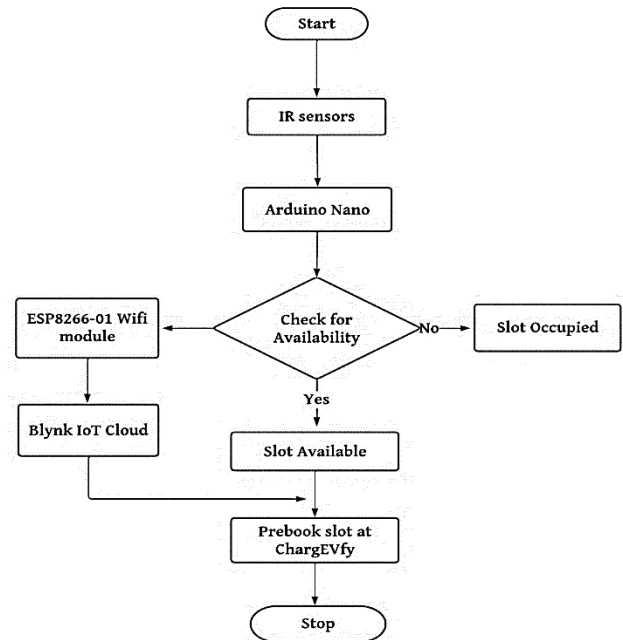


Fig. 16. Flowchart for Slot Availability

Fig.16 illustrates the flowchart for the slot availability and Fig.17 shows the hardware setup for slot availability system using IoT. The Arduino Nano has been utilized to control the LED's light transmits. The data from the sensor and LED is controlled by the Arduino Nano which will be collected by ESP8266 Wi-Fi module and transferred to the LCD display. The number of available slots and the position are displayed on the LCD display. The number '0' indicates the availability of the slot whereas, '1' indicates the unavailable state. The ESP 8266 Wifi module sends the slot availability data to the Blynk cloud over Wi-Fi, where it is available for the user to access and reserve the slot.

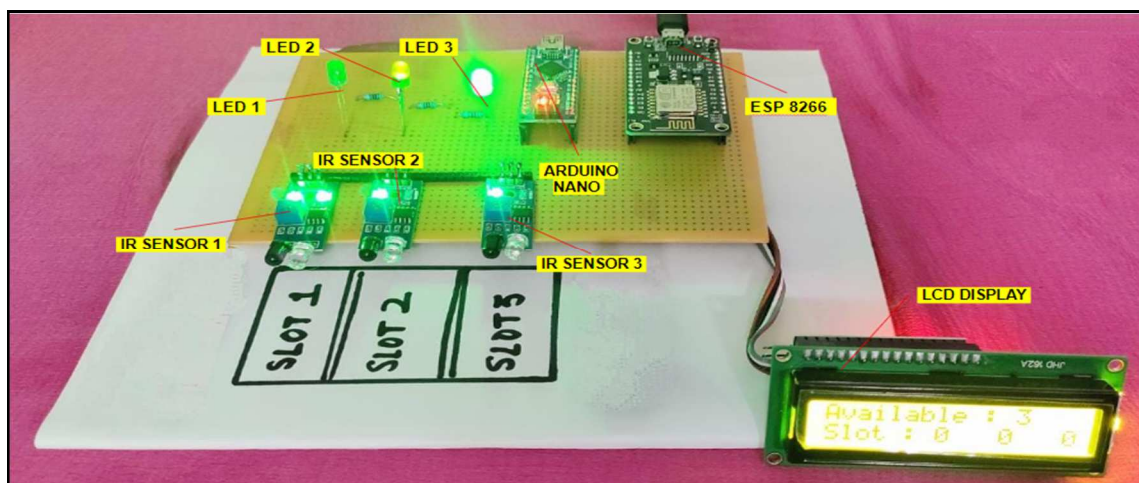


Fig. 17. Hardware Setup for Slot Availability System using IoT

VII. WEB BASED USER INTERFACE – CHARGEVfy

The name of the webpage is ChargeVfy. The webpage begins with a login form, which requests the user's username and password, as well as the registered EV in use. Once authorized, the user has access to several features that we believe will assist to alleviate EV concerns [10]. Moreover, the webpage provides options for (i) advanced slot booking option for charging of the user's EV, (ii) locating all the nearby charging stations around the user's surroundings, and (iii) monitoring of the user's electric vehicle.

A. Advanced Slot Booking option

The advanced feature of the developed webpage is that the customers can reserve a charging spot for their preferred time. The webpage displays every other hour details, such as whether or not the slot is reserved at the selected charging station. Three slots for each charging station timing display for a particular day is depicted in Fig. 18. The user must log in to the website, select the slot booking option, and then click on the desired timing. Every hour, the webpage informs the user whether the slot is booked or not. so that the user can locate a free slot. Following the selection of the preferred slot time, the user will be prompted to select the slot number.

SLOT 1	SLOT 2	SLOT 3
7:00 AM - 8:00 AM BOOKED	7:00 AM - 8:00 AM BOOKED	7:00 AM - 8:00 AM BOOKED
8:00 AM - 9:00 AM BOOKED	8:00 AM - 9:00 AM NOT BOOKED	8:00 AM - 9:00 AM BOOKED
9:00 AM - 10:00 AM NOT BOOKED	9:00 AM - 10:00 AM BOOKED	9:00 AM - 10:00 AM BOOKED
10:00 AM - 11:00 AM BOOKED	10:00 AM - 11:00 AM NOT BOOKED	10:00 AM - 11:00 AM NOT BOOKED
11:00 AM - 12:00 PM BOOKED	11:00 AM - 12:00 PM NOT BOOKED	11:00 AM - 12:00 PM NOT BOOKED
12:00 PM - 1:00 PM NOT BOOKED	12:00 PM - 1:00 PM NOT BOOKED	12:00 PM - 1:00 PM NOT BOOKED
1:00 PM - 2:00 PM BOOKED	1:00 PM - 2:00 PM NOT BOOKED	1:00 PM - 2:00 PM NOT BOOKED
2:00 PM - 3:00 PM NOT BOOKED	2:00 PM - 3:00 PM BOOKED	2:00 PM - 3:00 PM NOT BOOKED
3:00 PM - 4:00 PM NOT BOOKED	3:00 PM - 4:00 PM BOOKED	3:00 PM - 4:00 PM NOT BOOKED
4:00 PM - 5:00 PM NOT BOOKED	4:00 PM - 5:00 PM BOOKED	4:00 PM - 5:00 PM NOT BOOKED

Fig. 18. Timing selection option in webpage

The user can calculate their travel based on how much power is left in the vehicle by choosing the charging station

based on the distance the vehicle can cover. The user can pre-plan their travel and so that they do not have to wait in crowded charging stations. The user will then get confirmation by email or message based on the information provided in the sign in choice. Fig.19 illustrates the ChargeVfy website's slot booking option.

B. Locating all the nearby charging stations

The ChargeVfy website allows users to locate all charging stations along their route to their destination. The user must select the charging station location option on the homepage and enter the name of their intended destination to gain access to all charging stations in their radius. Fig.20 shows all of the charging stations along the route of the user's desired location.

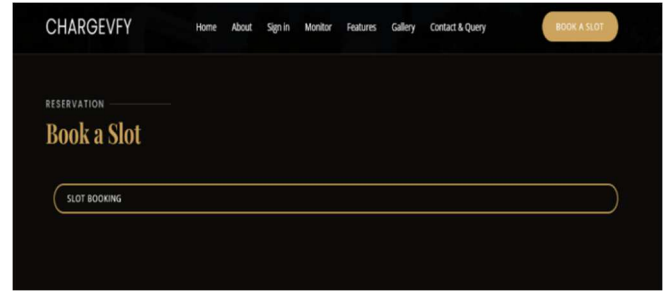


Fig. 19. Slot booking option in Webpage

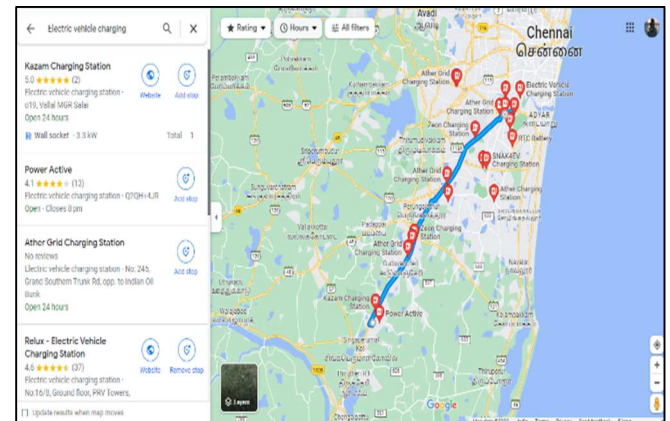


Fig. 20. Locating the charging stations

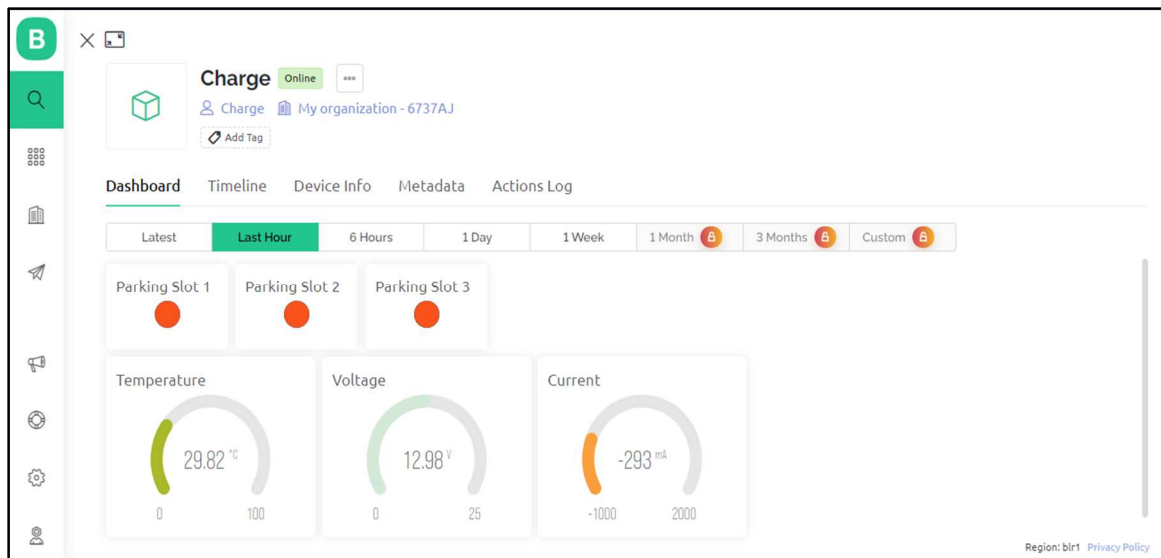


Fig. 21. Blynk Dashboard displaying all the sensor data

C. Monitoring of User's Vehicle

As shown in Fig.21 the user can monitor the availability of slots at the charging stations and the EV parameters like the battery voltage, current and temperature at the Blynk Console Dashboard which can be viewed at our web application.

VIII.CONCLUSION

Thus, IoT is used to power the link between the user and the electric vehicle. The idea presented in this paper reduces the amount of hassle experienced at charging stations. The user can make use of the data that is available about the battery of the EV through the sensors in order to take note of the state and healthiness of the EV battery in order to guarantee that the operation is carried out in a safe manner. The data that has been stored may be retrieved via the cloud using any software programs, such as Blynk, and with this information, a determination regarding whether to continue travelling or locate a charging station can be made. In order to further improve the system's dependability, a web-based user interface is also developed.

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